

PART 1 — LEVEL 1: Electrostatics

SECTION 1: Electric Charges & Properties (Q1–Q20)

1. Define electric charge.

Electric charge is a fundamental property of matter that causes it to experience a force in an electric field. It can be **positive** or **negative**, and is carried by subatomic particles: electrons (negative) and protons (positive).

2. State the SI unit of electric charge.

The SI unit of electric charge is the **coulomb (C)**.

3. What are the basic properties of electric charge?

- **Quantization:** Charge exists in discrete units of $e = 1.602 \times 10^{-19} \text{ C}$.
- **Conservation:** Total charge in an isolated system remains constant.
- **Additivity:** Total charge is the algebraic sum of individual charges.
- **Two types:** Positive and negative charges.

4. Explain “quantisation of charge.”

The quantisation of charge means that **charge exists in integer multiples of the elementary charge e** . That is, any charge $q = ne$, where n is an integer and $e \approx 1.602 \times 10^{-19} \text{ C}$.

5. Define “conservation of charge.” Give an example.

Conservation of charge states that **charge cannot be created or destroyed**, only transferred.

Example: When rubbing a glass rod with silk, electrons move from glass to silk, but the **total charge remains the same**.

6. What are positive and negative charges?

- **Positive charge:** Charge carried by protons.
- **Negative charge:** Charge carried by electrons.

7. What is meant by additive property of charges?

The **total charge** of a system is the **algebraic sum** of all individual charges.

$$q_{\text{total}} = q_1 + q_2 + q_3 + \dots$$

8. State the difference between like and unlike charges.

Property	Like Charges	Unlike Charges
Nature of force	Repel	Attract
Interaction	Same sign	Opposite sign

9. Can a charged body attract an uncharged body? Why?

Yes. Due to **electrostatic induction**, the uncharged body's charges rearrange so that opposite charges come closer, causing **attraction**.

10. What do you mean by charging by conduction?

Charging by conduction occurs when a **charged body transfers charge to a neutral body by direct contact**.

11. What do you mean by charging by friction?

Charging by friction occurs when **two different materials are rubbed together**, causing **electron transfer** from one material to another.

12. What is meant by charging by induction?

Charging by induction is **charging a body without direct contact**. A charged object is brought near a conductor, causing redistribution of charges. One part is grounded to remove excess charges.

13. Define an isolated charge.

An isolated charge is a charge **not influenced by other charges or conductors** nearby.

14. What is meant by point charge?

A point charge is an **idealized charge located at a single point**, with no spatial extent.

15. Explain free electrons.

Free electrons are electrons in **metallic conductors** that are **not bound to any atom** and can move freely under an electric field.

16. Why is the charge on an electron always the same?

Because **charge is a fundamental constant**, the electron always carries $-1.602 \times 10^{-19} \text{ C}$.

17. Name the instrument used to detect charge.

A **gold-leaf electroscope** or **pith-ball electroscope**.

18. What is a triboelectric series?

A triboelectric series is a list of materials **arranged by their tendency to gain or lose**

electrons when rubbed.

19. Why does a charged balloon stick to a wall?

Rubbing the balloon induces opposite charges on the wall surface. The attraction between opposite charges causes the balloon to stick.

20. Give two differences between positive and negative charges.

Property	Positive Charge	Negative Charge
Carrier	Proton	Electron
Mobility	Fixed	Highly mobile in conductors

SECTION 2: Conductors, Insulators & Induction (Q21–Q40)

21. What is a conductor? Give two examples.

A conductor allows free movement of electric charges.

Examples: Copper, Aluminum.

22. What is an insulator? Give two examples.

An insulator does not allow free movement of charges.

Examples: Rubber, Glass.

23. Define dielectric.

A dielectric is an insulating material that can be polarized in an electric field, reducing the field inside it.

24. State two differences between conductor and insulator.

Property	Conductor	Insulator
Charge mobility	High	Low
Example	Copper	Rubber

25. What are semiconductors?

Semiconductors have intermediate conductivity, which can be altered by doping.

Examples: Silicon, Germanium.

26. Explain the process of charge induction.

When a charged object is brought near a conductor, it repels or attracts free electrons,

causing **opposite charges on the near side** and **same charges on the far side**, without contact.

27. Why is induction preferred over friction for charging sensitive devices?

Induction **avoids direct contact**, preventing damage due to **high voltage sparks**.

28. What happens to charges on a conductor in equilibrium?

Charges **redistribute on the surface** so that the **electric field inside the conductor becomes zero**.

29. Why do charges reside on the surface of a conductor?

Because like charges **repel each other**, and maximum separation occurs on the **surface**.

30. Why is the electric field inside a conductor zero?

Charges redistribute to cancel any internal field; otherwise, free electrons would move until equilibrium is reached.

31. Define grounding.

Grounding is **connecting a charged object to the Earth** to neutralize it.

32. What is earthing?

Earthing is the same as grounding—**providing a path for excess charge to flow into Earth**.

33. Explain why metals are good conductors.

Metals have **many free electrons** that can move easily under an electric field.

34. What is polarization of charge in insulators?

Polarization is **slight separation of charges** within an insulator when an electric field is applied.

35. Why does a comb attract small pieces of paper after rubbing?

Rubbing charges the comb; the comb induces **opposite charges in paper**, causing attraction.

36. What is a hollow conductor?

A hollow conductor is a conductor with an **empty interior**, often used for shielding.

37. State the shielding effect of a conductor.

A conductor **blocks external electric fields**, protecting internal objects from external charges.

38. Why is a lightning conductor used?

To **safely discharge lightning** to the ground, preventing structural damage.

39. What is surface charge density?

Surface charge density (σ) is the **charge per unit area** on a conductor:

$$\sigma = \frac{Q}{A}$$

40. Why does a sharp metallic point leak charge easily?

Electric field at a sharp point is **very high**, causing charges to **escape into the air** (corona discharge).

✅ Up to this point, **Sections 1 and 2** are fully covered.

I can continue with **Sections 3–7** covering Coulomb's Law, Electric Field, Dipole, Flux, and Gauss's Law in the **next parts**, providing detailed explanations, formulas, and examples.

Do you want me to continue with **SECTION 3: Coulomb's Law** next?

yes

Perfect! Let's continue with **PART 1, LEVEL 1 — SECTION 3: Coulomb's Law (Q41–Q60)**. I'll provide **detailed explanations, formulas, and examples** for each.

SECTION 3: Coulomb's Law (Q41–Q60)

41. State Coulomb's law.

Coulomb's law states:

"The electrostatic force between two point charges is **directly proportional to the product of their magnitudes** and **inversely proportional to the square of the distance between them**."

42. Write the mathematical form of Coulomb's law.

$$F = k_e \frac{|q_1 q_2|}{r^2}$$

Where:

- F = electrostatic force
- q_1, q_2 = charges
- r = distance between charges
- $k_e = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2$

43. Define the SI unit of force in electrostatics.

The SI unit of force is **Newton (N)**.

44. What is the meaning of the term “electrostatic force”?

It is the **force exerted between stationary charges** due to their electric charges.

45. What is the value of Coulomb’s constant?

$$k_e = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2 \approx 9 \times 10^9 \text{ N m}^2/\text{C}^2$$

46. Define linearity of electric forces.

Linearity means that the **resultant force** on a charge is the **vector sum of forces from all other charges**, following the **superposition principle**.

47. On what factors does the electrostatic force depend?

1. Magnitude of charges (q_1, q_2)
2. Distance between charges (r)
3. Medium between the charges (via permittivity ϵ)

48. Does the electrostatic force follow Newton’s third law?

Yes. **Action = Reaction**: Forces on two charges are **equal in magnitude and opposite in direction**.

49. What is the nature of electrostatic force between like charges?

Repulsive. **Both charges push away from each other**.

50. What is the nature of electrostatic force between unlike charges?

Attractive. **Charges pull toward each other**.

51. Explain the concept of superposition principle.

The **net electrostatic force** on a charge is the **vector sum** of forces due to all other charges individually.

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \dots$$

52. Write the vector form of Coulomb's law.

$$\vec{F}_{12} = k_e \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

Where $\hat{r}_{12}$ is the unit vector from q_1 to q_2 .

53. Define medium permittivity.

Permittivity (ϵ) is the property of a medium that **affects the magnitude of electrostatic force** between charges:

$$F = \frac{1}{4\pi\epsilon} \frac{|q_1 q_2|}{r^2}$$

54. Explain relative permittivity.

Relative permittivity (ϵ_r) is the **ratio of permittivity of a medium to that of vacuum**:

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

55. Why is electrostatic force smaller in water than in air?

Water has a **high relative permittivity** ($\epsilon_r \approx 80$), reducing the force:

$$F = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{|q_1 q_2|}{r^2}$$

56. What is meant by a central force?

A central force acts **along the line joining two particles** and depends only on the distance between them (like Coulomb or gravitational force).

57. How is Coulomb's law similar to Newton's law of gravitation?

- Both inverse-square laws ($F \propto 1/r^2$)
- Both are central forces

58. How is Coulomb's law different from gravitational law?

Property	Coulomb's Law	Gravitational Law
Nature	Attractive or repulsive	Always attractive
Dependence	Depends on charge	Depends on mass
Medium effect	Yes, via ϵ	No, independent of medium

59. What are the limitations of Coulomb's law?

- Applies **only to point charges or spherically symmetric charge distributions**
- Does not account for **motion of charges** (electrodynamics)
- Medium is assumed **uniform and stationary**

60. What is effective force on a charge placed between two equal charges?

If two equal charges q are placed on either side of a test charge q_0 :

- **If charges are like:** Forces cancel → **net force = 0**
- **If charges are opposite:** Forces add → **net force = $2F$** toward the negative charge.

✅ **SECTION 3 is complete.**

Next, we can continue with **SECTION 4: Electric Field & Field Lines (Q61–Q90)**, which is crucial for understanding field concepts and applications.

Do you want me to continue with **Section 4 next?**

yes

Great! Let's continue with **PART 1, LEVEL 1 — SECTION 4: Electric Field & Field Lines (Q61–Q90)**. I'll provide **detailed explanations, formulas, and examples** for clarity.

SECTION 4: Electric Field & Field Lines (Q61–Q90)

61. Define electric field.

An electric field is a region around a **charged object** where it exerts a **force on other charges**. It is defined as **force per unit positive charge**:

$$\vec{E} = \frac{\vec{F}}{q_0}$$

62. Write the SI unit of electric field.

SI unit: **newton per coulomb (N/C)** or equivalently **volt per meter (V/m)**.

63. Give the mathematical expression for electric field due to a point charge.

$$E = k_e \frac{|Q|}{r^2}$$

- Q = source charge
- r = distance from charge

64. What is a test charge?

A **small positive charge** used to **measure the electric field** at a point. Its presence **does not disturb the existing field**.

65. What is meant by electric field intensity?

Electric field intensity is the **force experienced per unit positive charge** at a point in an electric field:

$$E = \frac{F}{q}$$

66. Define electric field lines.

Electric field lines are **imaginary lines showing the direction of the electric field** at every point.

67. State two properties of electric field lines.

1. They **start on positive charges** and **end on negative charges**.
2. Their **density indicates the strength** of the field (closer lines → stronger field).

68. Why do electric field lines never intersect?

If they did, the **field would have two directions at a single point**, which is impossible.

69. What is a radial field?

A radial field is one where **field lines radiate outward or inward from a point charge**, like around a single point charge.

70. What is a uniform electric field?

A uniform field has **same magnitude and direction** at every point, typically produced **between two parallel charged plates**.

71. Draw electric field lines for a positive point charge.

- Lines **radiate outward** from the charge.

72. Draw electric field lines for a negative point charge.

- Lines **converge inward** toward the charge.

73. Draw field lines for an electric dipole.

- Lines **emerge from positive charge** and **terminate on negative charge**.
- Lines **curve outward forming a dipole pattern**.

74. What is meant by tangential property of field lines?

The direction of the electric field at a point is tangent to the field line at that point.

75. Define the term "electric flux density."

Electric flux density $\vec{D}$ is the electric flux per unit area of a surface:

$$\vec{D} = \epsilon \vec{E}$$

Where ϵ = permittivity of the medium.

76. What is the physical significance of electric field?

Electric field represents the force a unit positive charge would experience at a point.

77. What is neutral point?

A neutral point is a point where net electric field is zero due to cancellation of fields from multiple charges.

78. What is the relation between force and electric field?

$$\vec{F} = q\vec{E}$$

Where q = charge experiencing the field.

79. Define electric field due to an infinitely long charged wire.

For a line charge with linear charge density λ :

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

- Radial outward for positive charge
- Radial inward for negative charge

80. State the expression for electric field due to a ring of charge on its axis.

For a ring of radius R and total charge Q , at distance x from center on axis:

$$E = \frac{kQx}{(x^2 + R^2)^{3/2}}$$

81. What is electric field inside a charged spherical shell?

Inside a spherical shell, electric field is zero ($E = 0$) at any point.

82. What is electric field outside a charged spherical shell?

Outside, the field behaves like a point charge at the center:

$$E = k\frac{Q}{r^2} \quad (r > R)$$

83. What is an electric monopole?

A monopole is a **single isolated charge** producing a radial field.

84. Why does electric field represent direction of force on a positive charge?

By convention, field lines point in the direction a positive test charge would move.

85. What is meant by electric field mapping?

It is **drawing field lines** or using **test charges** to visualize electric field patterns in a region.

86. What is space charge?

Space charge is a **distribution of charges in a region of space**, often in a vacuum or gas.

87. Can electric field lines be curved?

Yes. Field lines **curve** when multiple charges interact, e.g., **dipoles**.

88. What does closeness of field lines indicate?

Closer lines → **stronger electric field**.

89. What does divergence of field lines indicate?

Diverging lines → **weaker electric field**.

90. What is electric field due to a dipole at axial position?

Along the axis of a dipole ($r \gg d$):

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

Where $p = q \cdot 2a$ is the dipole moment, and r is distance from center.

✅ **SECTION 4 is complete.**

Next, we can move to **SECTION 5: Electric Dipole & Dipole Moment (Q91–Q110)**, which is crucial for understanding molecules, torque, and dipole behavior.

Do you want me to continue with **Section 5 next?**

yes

Perfect! Let's continue with **PART 1, LEVEL 1 — SECTION 5: Electric Dipole & Dipole Moment (Q91–Q110)**. I'll provide detailed explanations, formulas, and examples.

SECTION 5: Electric Dipole & Dipole Moment (Q91–Q110)

91. What is an electric dipole?

An electric dipole is a system of **two equal and opposite charges** ($+q$ and $-q$) separated by a distance $2a$.

92. Define dipole moment.

Dipole moment ($\vec{p}$) is a **vector quantity** defined as the **product of one of the charges and the separation distance**:

$$\vec{p} = q \cdot 2a$$

Direction: From **negative charge to positive charge**.

93. What is the SI unit of dipole moment?

SI unit: **Coulomb-meter (C·m)**

94. Give two examples of natural dipoles.

1. Water molecule (H_2O)
2. Hydrogen chloride molecule (HCl)

95. Write expression for electric field on axial line of dipole.

At a point on the axis at distance r ($r \gg a$):

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

Where $p = q \cdot 2a$.

96. Write expression for electric field on equatorial line of dipole.

At a point on the perpendicular bisector (equatorial line):

$$E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$$

Direction: **opposite to the dipole moment**.

97. What is the direction of dipole moment?

From **negative charge to positive charge**.

98. What is torque on a dipole in uniform electric field?

$$\tau = pE \sin \theta$$

Where θ is the angle between **dipole moment** and **electric field**.

99. When is torque on dipole maximum?

Torque is maximum when $\sin \theta = 1$, i.e., $\theta = 90^\circ$ (dipole perpendicular to field).

100. When is torque on dipole zero?

Torque is zero when $\theta = 0^\circ$ or 180° (dipole parallel or antiparallel to field).

101. What is the physical significance of dipole moment?

It measures the **strength of a dipole**, i.e., how much separation of charge exists. Larger p → stronger dipole.

102. Explain neutral point in dipole.

A neutral point is a **point along the axial line or equatorial line** where the **net electric field due to the dipole is zero**.

103. What is dipole in external electric field?

A dipole in an external field **experiences a torque** trying to **align it with the field**, but the **net force is zero in a uniform field**.

104. Define polarized molecule.

A polarized molecule has **separated positive and negative charges** due to an external electric field, forming an **induced dipole**.

105. What is nonpolar molecule?

A nonpolar molecule has **no permanent dipole moment** because the **charge distribution is symmetric**, e.g., O_2 , N_2 .

106. What is the behaviour of dipole in non-uniform electric field?

In a non-uniform field, a dipole experiences both:

- **Torque** (alignment)
- **Net force** (toward stronger field)

107. What is dipole approximation?

When **distance r from the dipole is much larger than separation $2a$ ($r \gg a$)**, the dipole can be treated as **point dipole**, simplifying calculations.

108. What is effective length of dipole?

The effective length $l_{\text{eff}} = 2a$, i.e., distance between the charges.

109. What do you mean by permanent dipole?

A permanent dipole is a molecule with a **natural, fixed dipole moment**, independent of external fields (e.g., water).

110. What is induced dipole?

An induced dipole is a **temporary dipole created** in a molecule due to an **external electric field**, even if the molecule is initially nonpolar.

✅ SECTION 5 is complete.

Next, we can continue with **SECTION 6: Electric Flux (Q111–Q125)**, which is foundational for Gauss's Law and field concepts.

Do you want me to continue with **Section 6 next?**

yes

Excellent! Let's continue with **PART 1, LEVEL 1 — SECTION 6: Electric Flux (Q111–Q125)**. I'll provide **detailed explanations, formulas, and examples**.

SECTION 6: Electric Flux (Q111–Q125)

111. Define electric flux.

Electric flux (Φ_E) is the **measure of the number of electric field lines passing through a given surface**.

$$\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \theta$$

Where:

- E = electric field magnitude
- A = area of surface
- θ = angle between field and area vector

112. Write the SI unit of electric flux.

SI unit: **Newton meter squared per Coulomb (N·m²/C)**

113. What is meant by solid angle?

A solid angle (Ω) is the 3D equivalent of plane angle, defined as:

$$\Omega = \frac{A}{r^2}$$

Where A = area on the sphere of radius r . Unit: **steradian (sr)**

114. What is differential flux area?

Differential flux area is a **very small surface element** $d\vec{A}$ over which the electric field is assumed uniform.

$$d\Phi = \vec{E} \cdot d\vec{A}$$

115. State the relation between electric field and flux.

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

- Electric flux is the **surface integral of electric field over an area.**

116. What is the significance of electric flux?

It **quantifies the field passing through a surface**, helping to calculate **enclosed charge** via Gauss's law.

117. When is electric flux maximum?

Flux is maximum when the field is **perpendicular to the surface** ($\theta = 0^\circ$):

$$\Phi_{\max} = EA$$

118. When is electric flux zero?

Flux is zero when the field is **parallel to the surface** ($\theta = 90^\circ$).

119. Define angle of dipole moment with area vector.

It is the angle θ between the **dipole moment vector** and the **area vector** of a surface when calculating flux.

120. What is area vector?

The area vector ($\vec{A}$) is a **vector perpendicular to the surface** with magnitude equal to the surface area:

$$|\vec{A}| = \text{area of surface}$$

Direction: **outward** for closed surface.

121. Write the formula for flux through a plane surface in uniform electric field.

$$\Phi_E = E \cdot A \cdot \cos \theta$$

Where θ is the angle between **field** and **area vector**.

122. What is electric flux through closed surface?

Electric flux through a closed surface (Gaussian surface) is the **total number of field lines leaving the surface**. Can be positive or negative depending on charge inside.

123. What is positive flux?

Flux is positive when **field lines leave the surface** (outward direction).

124. What is negative flux?

Flux is negative when **field lines enter the surface** (inward direction).

125. State any two properties of flux.

1. Flux is **proportional to the number of field lines** passing through the surface.
2. Flux is **maximum when field is perpendicular** to surface and zero when parallel.

✅ **SECTION 6 is complete.**

Next, we can continue with **SECTION 7: Gauss's Law & Applications (Q126–Q150)**, which ties together flux, fields, and charge distributions.

Do you want me to continue with **Section 7 next?**

yes

Perfect! Let's finish **PART 1, LEVEL 1** with **SECTION 7: Gauss's Law & Applications (Q126–Q150)**. I'll give **detailed explanations, formulas, and examples**.

SECTION 7: Gauss's Law & Applications (Q126–Q150)

126. State Gauss's law in electrostatics.

The total electric flux through a **closed surface** is equal to the **net charge enclosed** divided by the **permittivity of the medium**:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

127. Write mathematical form of Gauss's law.

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

128. What is Gaussian surface?

A **hypothetical closed surface** used to apply Gauss's law to calculate electric field, chosen **symmetrically** around charges.

129. What is symmetrical charge distribution?

A distribution where **field lines have regular patterns**, e.g.,

- Spherical
- Cylindrical
- Planar

130. Why is Gauss's law useful?

It simplifies calculation of electric field for **highly symmetric charge distributions** without using Coulomb's law for every point.

131. What is electric field for infinite line charge?

For line charge with **linear density** λ , at distance r :

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

132. State application of Gauss's law for infinite plane sheet.

For infinite sheet with **surface charge density** σ :

$$E = \frac{\sigma}{2\epsilon_0}$$

- Field is **uniform and perpendicular** to the plane.

133. State application of Gauss's law for spherical shell.

- **Outside shell:** $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
- **Inside shell:** $E = 0$

134. What is electric field inside a uniformly charged sphere?

$$E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} \quad (r < R)$$

- Linear increase with r inside.

135. What is electric field outside a uniformly charged sphere?

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (r > R)$$

- Behaves like a point charge at center.

136. Why Gaussian surface should be symmetric?

To ensure electric field is either parallel or perpendicular to the surface everywhere, simplifying flux calculation.

137. What is flux through a cylindrical surface?

For a line charge along axis:

$$\Phi = E \cdot 2\pi rL$$

Where L = cylinder length, r = radius.

138. State limitations of Gauss's law.

- Only gives magnitude of field easily for highly symmetric cases
- Does not directly give direction for irregular charge distributions

139. What is meant by non-conducting solid sphere?

A sphere of insulating material in which charge is uniformly distributed throughout its volume.

140. Plot E vs r graph for a charged spherical shell.

- Inside ($r < R$): $E = 0$
- Outside ($r > R$): $E \propto 1/r^2$

Graph: zero inside, hyperbolic decrease outside

141. Plot E vs r graph for a uniformly charged solid sphere.

- Inside ($r < R$): $E \propto r$
- Outside ($r > R$): $E \propto 1/r^2$

Graph: linear rise inside, hyperbolic decay outside

142. What is linear charge density?

$$\lambda = \frac{Q}{L}$$

- Charge per unit length of a wire

143. What is surface charge density?

$$\sigma = \frac{Q}{A}$$

- Charge per unit area of a surface

144. What is volume charge density?

$$\rho = \frac{Q}{V}$$

- Charge per unit volume

145. Define E-field for two infinite sheets.

For two parallel sheets with equal and opposite charges:

$$E = \frac{\sigma}{\epsilon_0} \quad \text{between sheets, zero outside}$$

146. Explain why field inside a hollow conductor is zero using Gauss's law.

- Gaussian surface inside conductor encloses no net charge $\rightarrow E = 0$

147. What is electric field due to charged cylinder?

For radius $r < R$ (inside):

$$E = \frac{\rho r}{2\epsilon_0} \quad (\text{linear})$$

For $r > R$ (outside):

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{hyperbolic})$$

148. What is flux through a cube containing a charge?

$$\Phi = \frac{Q}{\epsilon_0}$$

- Total flux depends only on enclosed charge, not shape.

149. What is flux through a cube when charge lies outside?

$$\Phi = 0$$

- No net flux through surface if charge is outside.

150. Why does Gauss's law not give direction of electric field?

- Law gives magnitude of total flux
- Direction depends on charge distribution and symmetry, which must be analyzed separately.